A network analysis of the emergence of spatial language and spatial categories in learners of an emerging sign language

Lexical labels can support the construction of categories, particularly for categories that are less perceptually salient (Gentner & Rattermann, 1991). Most studies illustrating such a relationship between language and cognition draw data from crosslinguistic comparisons (e.g., Regier & Kay, 2009; Majid et al., 2004) or child language acquisition (e.g. Gelman & Coley, 1991). Here we draw from language emergence to examine how a new, changing language can inform the way its learners construct categories from the world. We investigated the emergence of lexical forms for spatial relations in a new sign language in Nicaragua (Lengua de Señas Nicaragüense, LSN) by comparing three age cohorts representing its first three decades (C1, C2, C3; Table 1). Stimuli consisted of 18 images (Fig. 1d) of topological spatial relations including containment (e.g., *in*), support (e.g., *on*), and occlusion (e.g., *under*). Languages vary in how such spatial relations are differentiated and named (Bowerman, 1996). Signers first sorted these images based on perceived similarity of spatial relation. They then described each image. Given the early emergence of analogical depictions of spatial relations in LSN (Pyers et al., 2015) and the prevalence of such constructions in sign languages (Emmorey, 2001), we predicted that the earliest learners of LSN, C1, would produce primarily analogical depictions of spatial relations, while later cohorts (C2 and C3) would use lexical forms more frequently. If observed, we further expected that such a language difference would be associated with a difference in sorting strategies.

We found that while analogical forms dominate across all cohorts, the use of lexical signs to label spatial relations increased over cohorts, although not equally across spatial relation type. All cohorts used signs labeling containment, but signs labeling support and occlusion appear to have emerged only in later cohorts (Table 1). Network visualizations of each cohort's sorting strategies (iGraph; Figure 1) illustrate that all cohorts were sensitive to the category of occlusion (*under*), with all cohorts grouping those stimuli into a cohesive category. However, C1 signers grouped containment and support together, possibly sorting based on figure-ground contact. Sensitivity to figure-ground contact was not associated with any measured linguistic variable; even among C1 signers who rarely produced a lexical form to label "under," this category is robust. The networks of C2 and C3 signers depict their separation of containment from support relations; note further that the C3 signers appear to be differentiating full containment (e.g., money in bag, mouse in shoe) from partial containment. The separation of full containment as a category may be associated with the near exclusive use by all cohorts of a lexical sign for "in" with full but not partial containment (full containment = 92% (n=45); partial containment = 8% (n=4), a distinction observed in other languages (e.g., Johannes et al., 2015). Thus, the LSN sign for "in" seems to develop a core meaning of full containment (e.g., Landau, 2025), rather than a broader meaning of full or partial containment. Across all cohorts, partial containment items were often sorted with support items.

The early emergence of containment as a category and as a lexical form aligns with observations of early sensitivity to this category among infants (Hespos & Baillargeon, 2001) and the early acquisition of associated linguistic forms by children (Clark, 2004). Thus, the pattern observed in the emergence of spatial language in LSN likely reflects an interplay among universal sensitivities to some spatial relations, that are then progressively shaped as a spatial lexicon emerges.

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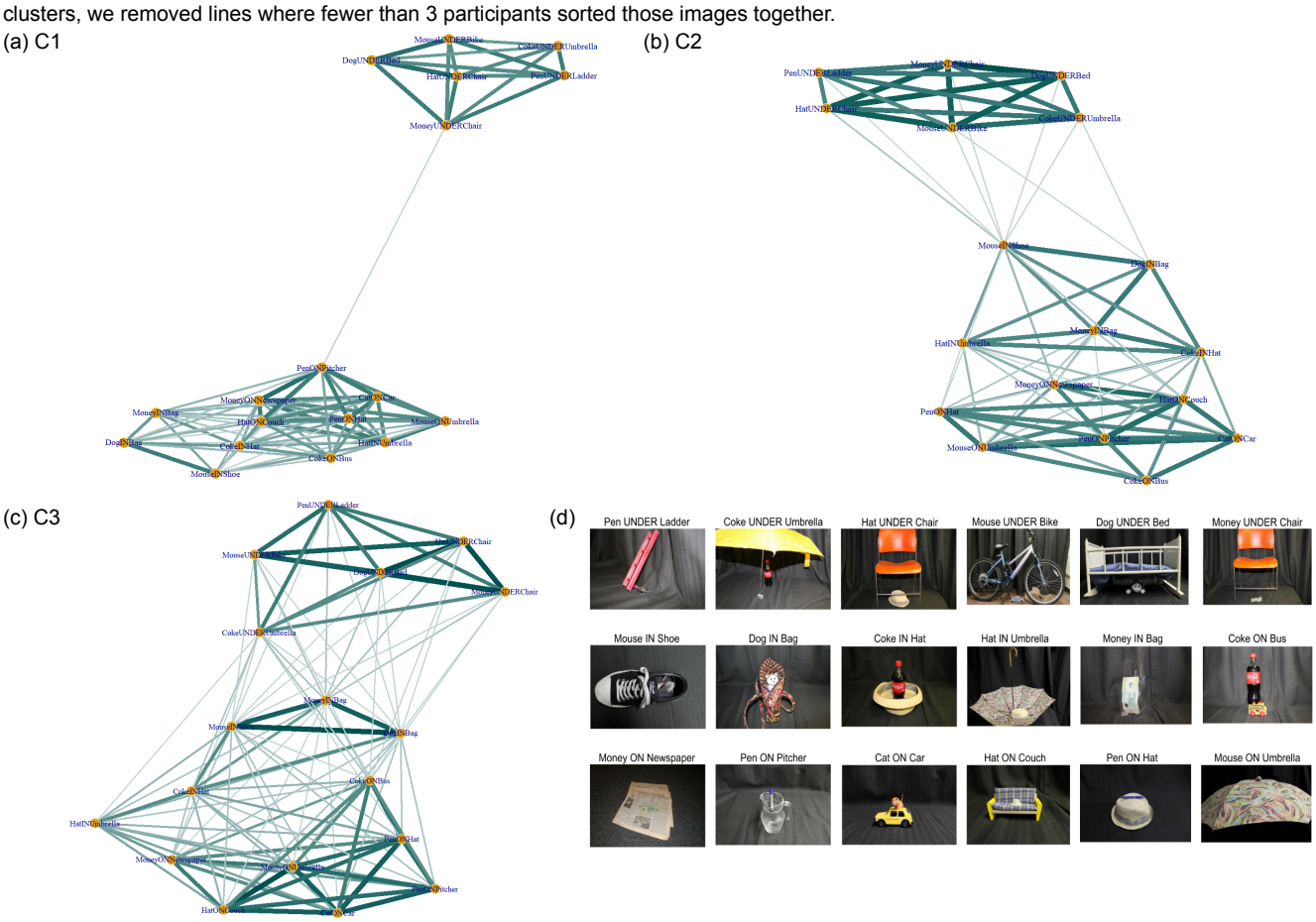
Table 1. Spatial language produced by LSN signers

		Analogical	Lexical			
		Total	Total	Containment (IN)	Support (ON)	Occlusion (UNDER)
Cohort	*N*	# tokens (# unique participants)	# tokens (# unique participants)	# tokens (# unique participants)	# tokens (# unique participants)	# tokens (# unique participants)
C1	12	156(12)	31(8)	17(8)	7(3)	7(3)
C2	13	190(13)	58(12)	19(8)	20(8)	19(8)
C3	12	188(12)	52(11)	13(6)	19(8)	20(8)

Figure 1. Network visualization of sorting strategies for (a) first (b) second, and (c) third cohort signers. Reference images provided in (d).

Color darkness and line thickness illustrate the number of participants who sorted images together. To enhance visibility of the clusters, we removed lines where fewer than 3 participants sorted those images together.